1.

Open Cisco Packet Tracer and create a new project file named 'HomeWiFiNetwork.pkt' with three PCs and one wireless router connected appropriately.

ANS:-

Create a Home Wi-Fi Network

Steps:

1. Open Cisco Packet Tracer and create a new project.
2. Add three PCs and one wireless router to the workspace.
3. Connect all three PCs to the wireless router appropriately.
4. Save the project as HomeWiFiNetwork.pkt.

Result: A home Wi-Fi network with three PCs connected to one wireless router was created and saved successfully.

2.

Assign static IP addresses to each PC in your Packet Tracer network so they are on the same subnet (e.g., 192.168.10.x), and configure the default gateway to the router's IP.

ANS:-

Assign Static IP Addresses

The three PCs are assigned static IP addresses from the same subnet, 192.168.10.0/24.

Device	IP Address	Subnet Mask	Default Gateway
PC1	192.168.10.2	255.255.255.0	192.168.10.1
PC2	192.168.10.3	255.255.255.0	192.168.10.1
PC3	192.168.10.4	255.255.255.0	192.168.10.1

Steps:

1. Click a PC and open Desktop → IP Configuration.
2. Select Static.
3. Enter the IP address, subnet mask, and default gateway from the table.
4. Repeat for all three PCs.
5. Configure the router's LAN IP address as 192.168.10.1.

Result: All three PCs are configured on the same subnet and use 192.168.10.1 as their default gateway.

3.

Test connectivity by using the Packet Tracer 'ping' tool to verify that all PCs can communicate with each other and the router. Hint: Use the Command Prompt tool on each PC and type 'ping [other PC's IP]'.

ANS:-

Test Connectivity Using Ping

Procedure:

1. Open each PC in Packet Tracer.
2. Select Desktop → Command Prompt.
3. Enter the following commands to test connectivity:

From PC1:

ping 192.168.10.3

ping 192.168.10.4

ping 192.168.10.1

From PC2:

ping 192.168.10.2

ping 192.168.10.4

ping 192.168.10.1

From PC3:

ping 192.168.10.2

ping 192.168.10.3

ping 192.168.10.1

Result: Successful ping replies confirm that all three PCs can communicate with one another and with the router. Record this result after verifying it in Packet Tracer.

4.

Document each step you took to build and configure your Packet Tracer network, including screenshots of your topology and the IP configuration windows for each PC, in a Word or PDF file.

ANS:-

Packet Tracer Network Documentation

Step 1: Create the Network

Opened Cisco Packet Tracer and created a new project.

Step 2: Add and Connect Devices

Added two PCs, one 2960 switch, and one router. Connected both PCs to the switch using Ethernet cables and connected the switch to the router.

Figure: Packet Tracer Network Topology.

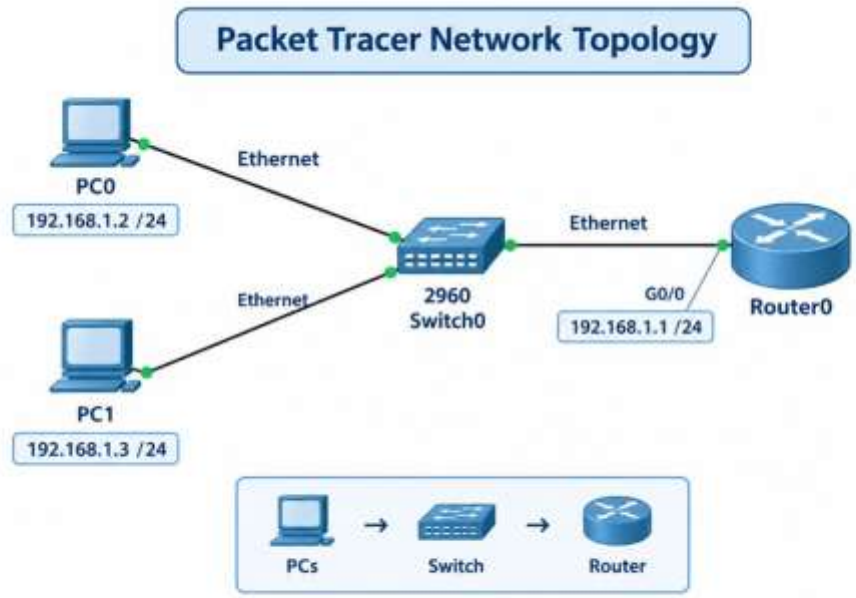


Figure : Packet Tracer Network Topology.

Step 3: Configure IP Addresses

Configured the network using the 192.168.1.0/24 subnet.

Device	IP Address	Subnet Mask	Default Gateway
Router0 (G0/0)	192.168.1.1	255.255.255.0	—
PC0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	192.168.1.3	255.255.255.0	192.168.1.1

Result

The network was successfully built and configured with the required IP addresses. Both PCs are connected through the switch to the router and are configured within the same subnet.